Insertion Sort with Duplicate Elements

1. Aim

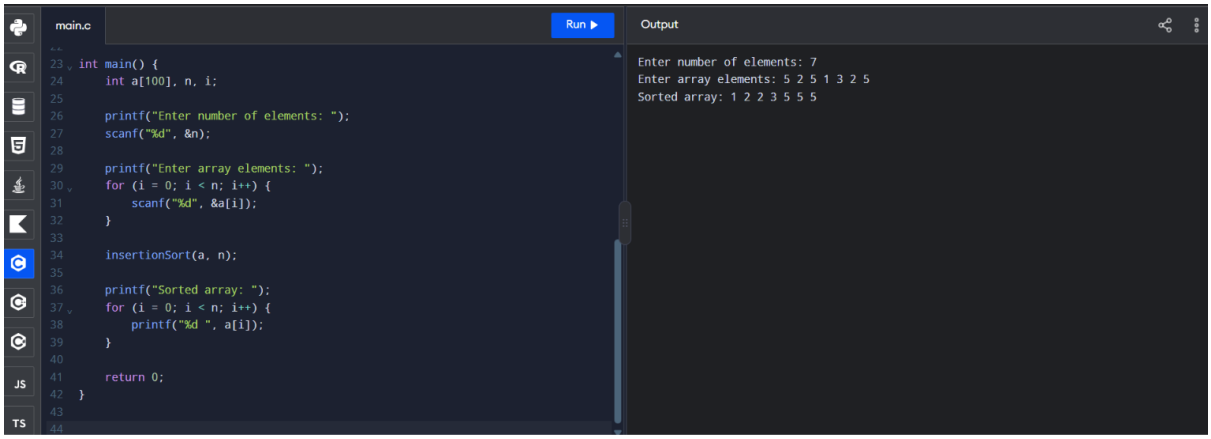
To implement Insertion Sort in C and analyze its performance when the array contains duplicate elements.

2. Theory

Insertion Sort is a sorting algorithm that builds a sorted array one element at a time. It takes each element and inserts it into its correct position among the previously sorted elements.

The algorithm starts with the second element and compares it with the elements before it. Larger elements are shifted one position to the right, and the selected element is inserted into the correct position.

Duplicate elements are handled correctly because equal elements are not shifted past one another when the comparison uses the greater-than operator.



5. Complexity Analysis

Case	Time complexity
Best case	\(O(n)\)
Average case	\(O(n^2)\)
Worst case	\(O(n^2)\)

Auxiliary space	\(O(1)\)

- Best case: The array is already sorted.
- Average case: Elements are in a mixed order.
- Worst case: The array is sorted in reverse order.

Result: Insertion Sort successfully sorts an array containing duplicate elements in ascending order.